Topic 23 — Disaster & Environment (पर्यावरण)

Covers syllabus (click to expand)

Floods | Drought (सूखा) | Landslides | Cyclones | Heat Waves | Forest Fires

Sources baked in: NCERT Geo 11–12; NDMA; IMD; Sendai Framework 2015; FSI; MoEFCC; UPPCS PYQs 2018–2025 **Weight:** ★★★ High — disaster–environment chain; UP flood/drought/heat; DPAP year; Piral Lao **Last verified:** August 2026

Current Affairs (this topic)

- Flood / landslide / heat-wave seasons refresh disaster–environment CA for UP–Himalaya (हिमालय).
- Sendai Framework year (**2015**) stays fixed — link hazards to land-use and climate stress.

23.1 Floods

Identity: Riverine / flash / urban | UP Ganga–Yamuna–Terai | NDMA 2005 | Sendai 2015

- A **flood** occurs when water overflows onto normally dry land — river channel capacity or drainage is exceeded.
- **Riverine floods** swell slowly over days–weeks on plains. **Flash floods** follow cloudbursts in hills within hours. **Urban flooding** is waterlogging from impervious surfaces and clogged drains.
- **UP flood rivers:** Ganga, Yamuna, Ghaghara (Saryu), Rapti, Gandak, Gomti** — monsoon inundation of the Gangetic floodplain.
- **Human amplification:** floodplain encroachment, embankments that raise silted beds, and **deforestation** in catchments that speeds runoff.
- **Terai flash risk:** Himalayan/Nepal-side catchments can inundate **Pilibhit, Lakhimpur Kheri, Bahraich** rapidly after extreme rain.
- **Natural flood benefit:** controlled flooding deposits **alluvial (जलोढ़) silt** that renews soil fertility — the problem is uncontrolled flooding of encroached zones.
- **Institutions:NDMA (Disaster Management Act 2005)** — **Prime Minister** is Ex-officio Chair; **NDRF** for response; **IMD/CWC (केंद्रीय जल आयोग)** for forecasting; **Sendai Framework 2015** for disaster risk reduction.
- **Mitigation:** floodplain zoning, wetland restoration (natural sponges), catchment afforestation, early warning.

Type	Speed	UP example
Riverine	Days–weeks	Ganga at Prayagraj (प्रयागराज)/Varanasi (वाराणसी)

Type	Speed	UP example
Flash	Hours	Terai foothills
Urban	Hours	Lucknow (लखनऊ), Kanpur (कानपुर) waterlogging

> **Logic:** Floods are India's most frequent disaster by affected population (जनसंख्या) — not limited to coastal states.

23.2 Drought

Identity: Three types | Bundelkhand hotspot | DPAP 1973 | Creeping (slow-onset) disaster

- **Drought** develops slowly — weeks to months — unlike sudden floods.
- **Meteorological drought:** rainfall below long-term normal.
- **Agricultural drought:** soil moisture insufficient for crops (even if some rain occurs).
- **Hydrological drought:** rivers, lakes, and groundwater below normal after prolonged dry spell.
- Sequence often runs meteorological → agricultural → hydrological.
- **Bundelkhand (UP):** seven districts — **Jhansi (झांसी), Lalitpur, Jalaun, Banda, Hamirpur, Mahoba, Chitrakoot (चित्रकूट)** — chronic drought–desertification (मरुस्थलीकरण) hotspot.
- **Drought Prone Areas Programme (DPAP)** launched in **1973** — soil conservation, water harvesting, afforestation in dryland districts.
- Chronology fact s: **CDP 1952** → **DPAP 1973** → **Food for Work 1977** → **TRYSEM 1979**.
- **Feedback loop:** drought kills vegetation → soil exposed → erosion → worse water retention → next drought worse.
- Other schemes: Watershed Development, **MGNREGA** water works, Bundelkhand Package, Jal Jeevan Mission.

PYQ — UPPCS Prelims 2025, Q134

Arrange chronologically: (1) Food for Work (2) Community Development Programme (3) TRYSEM (4) Drought Prone Areas Programme

A. 4, 2, 3, 1

B. 2, 4, 1, 3

C. 2, 4, 3, 1

D. 4, 2, 1, 3

Show answer

Ans: B — CDP **1952** → DPAP **1973** → FWP **1977** → TRYSEM **1979**.

> **Logic:** DPAP = **1973**, not 1952 (that is CDP).

23.3 Landslides

Identity: Gravity mass wasting | Himalaya/NE | Kedarnath 2013 | Human triggers matter

- A **landslide** is downhill movement of rock, soil, or debris when gravity overcomes slope stability.
- Needs **steep slope + weak geology + water lubrication + loss of root binding**.
- Types include slides, rock falls, **debris flows** (most deadly after cloudburst), and rotational slumps.

Kedarnath 2013 — Cause, Course, Result

Cause: Extreme rainfall / cloudburst over fragile Himalayan slopes, plus prior slope weakening from roads, settlement, and vegetation stress, set up a mass-wasting cascade near Kedarnath (Uttarakhand).

Course: Cloudburst triggered landslides and debris flows that choked and then released flash flood water through the Mandakini (मंदाकिनी) corridor — a hydrometeorological + mass-movement chain, not a pure earthquake or tsunami (सुनामी) frame.

Result: Catastrophic loss of life and infrastructure; became the national teaching case for environment–disaster coupling in the Himalaya. Downstream landslide-dam and flood risk also matters for Terai plains including UP.

- **Human triggers:** deforestation (roots bind soil), road cutting/blasting, unplanned hillside construction, mining — landslides are **not purely natural**.
- **Earthquake link:** Himalayan seismic shaking can trigger widespread mass failures.
- **Downstream risk:** landslide debris can dam (दाम -) rivers; dam-break floods then hit plains including UP Terai.
- **Mitigation:** bio-engineering, retaining walls, afforestation, land-use zoning on steep slopes, GSI hazard zonation maps.

> **Logic:** "Landslides are purely natural — humans play no role" = **FALSE**.

23.4 Cyclones

Identity: Warm ocean ≥26–27°C | Bay of Bengal nursery | Eye calm / eyewall strongest | Storm surge

- A **tropical cyclone** is an intense rotating storm over warm tropical oceans with a low-pressure centre.
- Formation needs **sea surface temperature ≥26–27°C**, moist air, low vertical wind shear, and Coriolis force (not on the equator).
- **Eye:** calm, clear, low-pressure centre. **Eyewall:** ring of **strongest winds and rain** around the eye. Trap: strongest winds are **not** in the eye.
- **Storm surge:** cyclone winds push seawater onshore — often deadlier than wind on flat coasts (Odisha, WB, Bangladesh).
- **Bay of Bengal** produces **more frequent and intense** cyclones than the **Arabian Sea** (warmer, shallower, moister waters).

- East-coast states most hit: Odisha, West Bengal, Andhra Pradesh, Tamil Nadu (नाडु). UP gets rain remnants, not landfall.
- Regional names: **cyclone** (Indian Ocean), **hurricane** (Atlantic), **typhoon** (NW Pacific), **Willy-Willies** (Australia), **Baguios** (Philippines).
- **Mangroves** (e.g. Sundarbans (सुंदरबन)) absorb surge energy — coastal buffer. Deforestation of mangroves increases damage.
- Recent named storms: **Amphan (2020)**, **Fani (2019)**, **Hudhud (2014)**, **Yaas (2021)** — IMD names North Indian Ocean cyclones.

PYQ — UPPCS Prelims 2018 pattern

Natural disasters cause max damage in developing countries; Bhopal was man-made; India is disaster-free; mangroves reduce cyclone impact.

Show answer

True statements: 1, 2, and 4. India is **not** disaster-free. Mangroves **do** reduce cyclone impact.

Feature	Eye	Eyewall
Winds	Calm	**Strongest**
Pressure	Lowest	Rising outward

> **Logic:** Baguios–Philippines; Hurricanes–USA; Typhoons–China/NW Pacific; Willy-Willies–Australia.

23.5 Heat Waves

Identity: IMD thresholds | Urban heat island | May–June peak | Climate frequency rise

- A **heat wave** is a prolonged period of abnormally high temperature relative (सापेक्ष) to the location's normal.
- **IMD plains criteria:** maximum temperature **≥40°C** with departure **+4.5°C to +6.4°C** = heat wave; departure **≥+6.4°C** or absolute (निरपेक्ष) temp **≥47°C** = **severe heat wave**.
- **Urban heat island:** cities stay hotter than surrounding rural areas due to concrete, asphalt, waste heat, and lost greenery — amplifies heat-wave mortality.
- UP hotspots: **Kanpur, Lucknow, Agra, Ghaziabad, Noida** — peak stress in **May–June** pre-monsoon.
- **Climate link:** warming makes heat waves more frequent, longer, and hotter (IPCC (आईपीसीसी) AR6).
- **Compound risk:** heat + drought raises crop failure and forest-fire fuel dryness; heat + power surge risks blackouts.
- **Health:** heatstroke, dehydration — elderly, outdoor workers, children most vulnerable.
- **Adaptation:** cool roofs, urban greening, IMD heat warnings, Heat Action Plans (Ahmedabad (अहमदाबाद) model).

Feature	Heat wave	Urban heat island
Scale	Regional weather event	Local city microclimate
Duration	Days–weeks	Persistent year-round bias
Cause	Atmospheric pattern	Built surfaces, lost greenery

> **Logic:** Heat wave ≠ urban heat island — different scale and cause.

23.6 Forest Fires

Identity: Chir pine fuel | March–June peak | Piral Lao–Paise Pao | FSI satellite alerts

- A **forest fire** is uncontrolled fire through forest vegetation — needs **fuel + oxygen + ignition**.
- **Surface fire** burns litter/grass; **crown fire** burns canopy (most destructive); **ground fire** smoulders in organic soil.
- **Chir pine** forests (Uttarakhand, Himachal (हिमाचल)) accumulate **dry pine needles** — major summer fuel load.
- Peak season: **March–June** before monsoon rains.
- Human causes dominate: pasture burning, negligence, arson — not only lightning.
- **Environmental damage:** biodiversity loss, soil organic matter destruction, erosion, **PM2.5 smoke**, CO_2 release; post-fire slopes raise landslide/flood risk.
- **FSI** uses satellite (MODIS/VIIRS) for real-time forest fire monitoring and state alerts.
- **"Piral Lao – Paise Pao"**: Uttarakhand campaign — villagers collect pine needles (**piral**), reducing fire fuel and earning income; launched under Uttarakhand CM.

PYQ — UPPCS Prelims 2024, Q86

Piral Lao–Paise Pao: (1) reduces forest fires and gives income to villagers (2) launched by Uttarakhand CM

A. Only 2

B. Neither

C. Only 1

D. Both 1 and 2

Show answer

Ans: D — Both statements correct — fire prevention + livelihood; Uttarakhand CM campaign.

> **Logic:** Piral Lao = **Uttarakhand** (उत्तराखंड) pine-needle campaign — not a UP state flagship scheme.
